Figure 1: SQLiteStudio Download

- **Plugins:** SQLite Studio supports plugins of different categories. For example, another script language could be added along with it.

What's a database?

Data can be facts related to any object. For example, your name, address, salary, date of birth, etc, is some of the data related to you. A picture, image, file, etc, can also be considered data.

An SQL Server database consists of a collection of tables that store a specific set of structured data. As an example, an electrical service provider uses a database to manage billing, customer issues and their addresses. A phone book uses a database to store peoples' names, phone numbers, contact details and addresses.

To create a database, click on *Database > Add Database (Ctrl + O) > New_Database*

Table: Data is stored in database objects called tables in RDBMS. This table is a collection of related data items, and consists of rows and columns. It is the most common and simplest form of storing data in a RDBMS.

An example of a 'Customer' table is given in Figure 3.

Row: A row (records or tuple) in a table represents a single, implicitly structured data.

Column: A column (or field) in a table is a vertical entity that contains all the information associated with a specific field. For example, a column in the 'Customer' table is *NAME*, which represents the names of all the customers

'Null' value: A 'null' value in a table is a value that appears empty. It doesn't mean 'zero' value but just the lack of a value.

SQL constraints

Constraints are the rules that apply to the data columns in a table. They are used to limit the type of data that can be included in a table. This ensures the accuracy and reliability of the database data.

Constraints can be at column or table level. Column-level restrictions apply to a single column, while table-level restrictions apply to the entire table.

- *Not null:* Ensures that the column cannot have a null value.
- *Primary key:* Uniquely identifies each row in a database table.
- *Foreign key:* Uniquely identifies a row/table in any other database table.
- *Unique:* Ensures that all values in a column are different.
- *Check:* Ensures that all values in a column satisfy certain conditions.

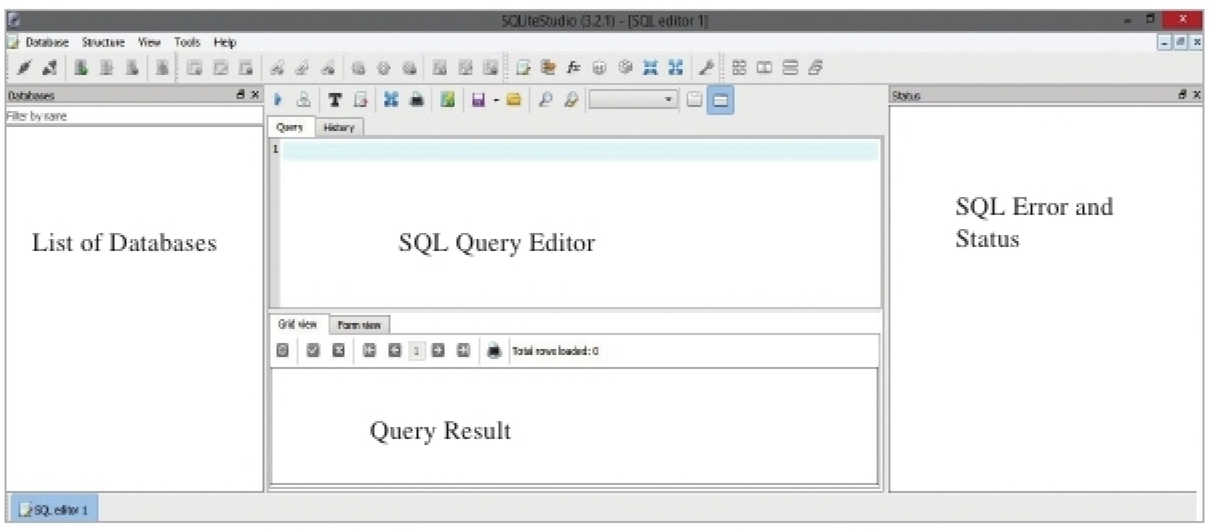

Figure 2: New database

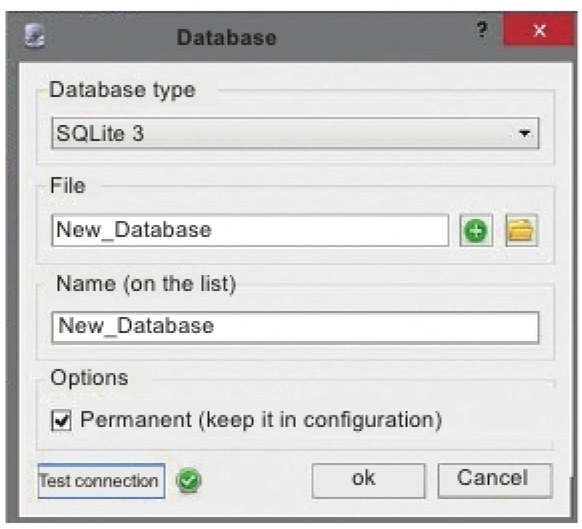

Figure 3: 'Customer' table